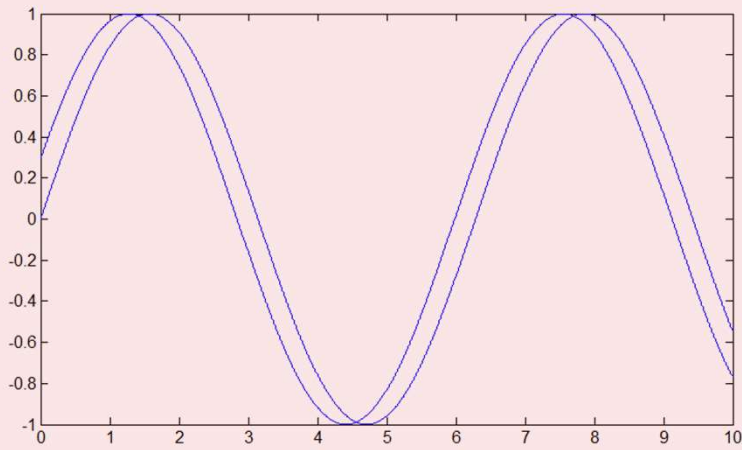
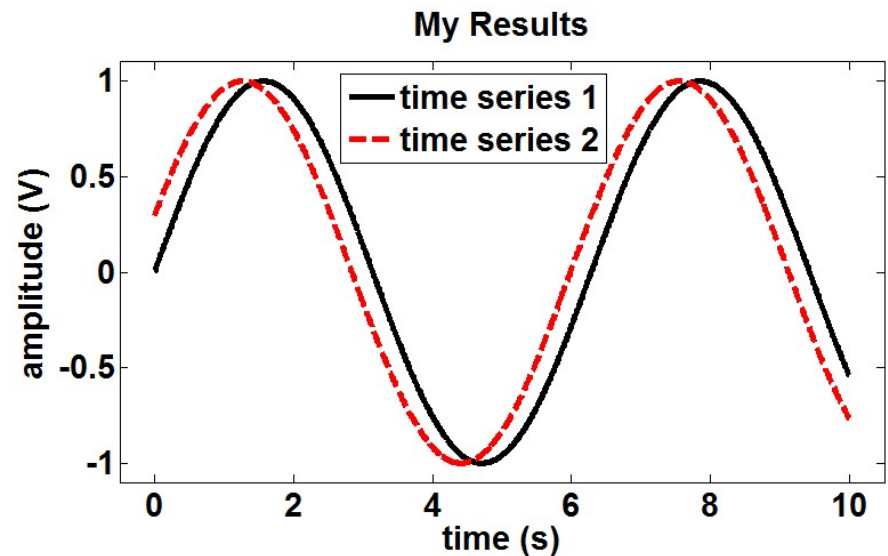


MATLAB Figure Example



LAZY-PERSON'S MATLAB PLOT – never submit this!

- No **axes** or plot **titles**
- Fonts **too small** to read
- Two data series are indistinguishable and no legend
- Inappropriate axes intervals



```
plot(x, y, 'k', 'LineWidth', 4);
hold on;
plot(x, y2, 'r--', 'LineWidth', 4);
xlim([-0.5, 10.5]);
ylim([-1.1, 1.1]);
xlabel('time (s)');
ylabel('amplitude (V)');
legend('time series 1', 'time series 2');
title('My Results');
% Set font for axes
set(gca, 'FontSize', 22, 'fontWeight', 'bold')
% Set font for all other text in figure
set(findall(gcf, 'type', 'text'), 'FontSize', 22, 'fontWeight', 'bold')
```

Inserting Figures in Reports

We will use two tumor models with known differences in biology and drug response. The PC3 line is androgen-independent metastatic prostate cancer cell line, and the PC3/2G7 is stem cell-like clone of PC3; both lines have been well characterized by us (Waxman Lab) in the context of cytotoxic therapy response³⁷. These two lines have been shown to create tumors with markedly different levels of vascularization, tumor blood flow, tumor cell drug uptake and drug-induced apoptosis. Male SCID Hairless mice will be inoculated on each flank (2 implantation sites/mouse). While we recognize the benefits of orthotopic models, SFDI and ESS measurements are greatly facilitated by the subcutaneous location compared to the deeper orthotopic prostate site. Additionally, our (Waxman Lab) preliminary studies establish that the vascular properties and differential therapeutic responsiveness of these tumor models are consistent in both subcutaneous and orthotopic implementations. Cytotoxic agents: For our first experiments, mice will be treated with a single dose of the alkylating agent cyclophosphamide (140 mg/kg) based on our prior experience with this drug in this and other tumor models^{3,102,103}, or vehicle control. Follow-up studies will be carried out with select other drugs that also induce apoptosis but act via different mechanisms. These may include TRAIL (500 µg/ 30 g mouse), which activates the extrinsic apoptosis pathway; cisplatin (5 mg/kg), a platinating agent; gemcitabine, a nucleoside analog and anti-metabolite (160 mg/kg); and docetaxel, a microtubule inhibitor (7.5 mg/kg)^{104,105}. Titrations: For one or two of the drugs, optical measurements will be taken on mice treated at different doses of each drug (e.g., 35, 140 and 280 mg/kg cyclophosphamide) to allow correlation of optical signal strength with degree of apoptosis.

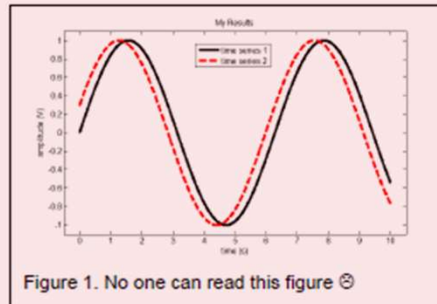


Figure 1. No one can read this figure ☹

Figure size appropriate, but font size too small

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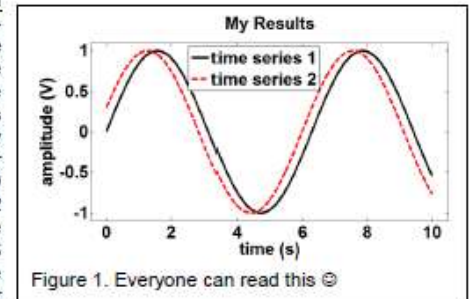


Figure 1. Everyone can read this ☺

Make font size huge in MATLAB (e.g., 32).
It will feel weird at first in MATLAB but will
integrate better into your report

Additional Resources

- General MATLAB plotting examples

<http://www.mathworks.com/help/matlab/ref/plot.html>

- Making pretty graphs

<http://blogs.mathworks.com/loren/2007/12/11/making-pretty-graphs/>

